

Engineering Assessment — Recovery Boom Weld Defect (Confidential)

Reference ENG-CONF-905 Revision A Unit: Engineering Branch

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1.2 Defect Rate

Non-destructive testing of the recovery boom heel weld identified a defect rate of 4.7 percent across manufacturing batch LOT-2291. The defects are sub-surface lack-of-fusion indications at the heel-to-pedestal joint.

2.3 Interim Operating Restriction

Pending completion of re-inspection, an embargo is placed on all boom lifts exceeding 6,500 kgf on vehicles drawn from batch LOT-2291. Recovery by winch is unaffected by this embargo.

3.1 Affected Population

Batch LOT-2291 comprises 46 boom assemblies, of which 31 are known to be in service. The remaining 15 are held in the spares pipeline and have been quarantined pending inspection.

4.4 Re-inspection Schedule

All in-service assemblies from the affected batch shall be re-inspected by ultrasonic method within 120 days. Assemblies passing re-inspection are released from the embargo without further restriction.

5.2 Root Cause

Root cause is assessed as an out-of-tolerance pre-heat temperature during manufacture, below the specified 120 degrees Celsius minimum. The manufacturing process has since been corrected.

6 Handling

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